

Florian Wiesner

www.flowsnr.github.io

EXPERIENCE

- **Mistral AI** June 2026 – Now
Applied AI Scientist & Research Engineer London, UK
- **University of Virginia** Feb. 2025 – April 2026
Visiting Scholar at the [Visual Intelligence Laboratory](#) Charlottesville, VA, United States
 - **Multi-modal AI for Science:** Research on multi-modal datasets and models (physics & language)
 - **Physics Foundation Model:** Developed a Physics Foundation Model capable of modeling unseen physical systems purely from previous context.
 - **Research supervision:** Supervised research of two PhDs regarding Graph Neural Networks and Autoencoders.
 - **Student mentoring:** Mentored PhD students, fostering software engineering best practices and presentation skills.
- **RWTH Aachen University** May 2021 – Feb. 2026
PhD, Researcher Aachen, Germany
 - **Group Leader TriggerINK Robotics:** Led the [TriggerINK](#) Robotics research team (5 PhDs) focused on robotic in-wound 3D-printing of an interactive bio-ink for supporting cartilage regeneration.
 - **Object Tracking with Synthetic Images:** Developed a mask-RCNN model trained on synthetic microscope images (rendered using Blender) to track microgels. Evaluated microgel flow using SORT and multi-hypothesis tracking algorithms.
 - **Thesis Supervision and Mentorship:** Managed 20+ undergraduate and graduate students and 5+ research assistants, fostering technical and personal development.
 - **Research Cooperation with University of Alberta:** Initiated a collaborative research effort between RWTH and the [ESDLab](#) at the University of Alberta, led by Prof. Marc Secanell.
 - **Research Project Management:** Managed a research project which was part of the [H2Giga](#) flagship project. The project had an investment budget of 500k Euro, a yearly budget of 10k Euro and a duration of 4 years.
 - **Teaching and Lectures:** Delivered lectures and tutorials for 3 years, enhancing student engagement.
 - **Lab Management:** Managed a large research laboratory (wet-lab) with around 10 active daily users.

EDUCATION

- **Dr.-Ing. (PhD) in Machine Learning & Chemical Engineering** May 2021 – Feb. 2026
RWTH Aachen University Aachen, Germany
Summa cum laude

PUBLICATIONS

- **Towards a Physics Foundation Model** July 2026
ICML AI4Physics 2026, [blog post](#), [arxiv](#)
- **Enhancing Bubble Removal in Geometry-Optimized Electrodes** May 2026
Advanced Engineering Materials, doi: [10.1002/adem.70921](#)
- **Data Mining for Enhanced PEM Electrolysis** Jan. 2026
Journal of The Electrochemical Society, doi: [10.1149/1945-7111/ae335e](#)
- **Modeling of Gas Diffusion Electrodes** Jun. 2024
ACS Applied Materials & Interfaces, doi: [10.1021/acsami.4c04641](#)
- **Electrowetting and Reactions in Microfluidic Gas Diffusion Electrode** Feb. 2024
Small, doi: [10.1002/sml.202310427](#)
- **Additive Manufacturing of Intertwined Electrode Pairs** Oct. 2022
Advanced Engineering Materials, doi: [10.1002/adem.202200986](#)

PERSONAL PROJECTS - [GITHUB](#)

- **Physlib:** Contributing to formalizing physics in lean.
- **ML-training suite:** Boilerplate pytorch code for DDP, compile, amp, checkpointing, time keeping on HPCs.
- **CUDA kernel:** Custom CUDA kernel for masked convolutions.